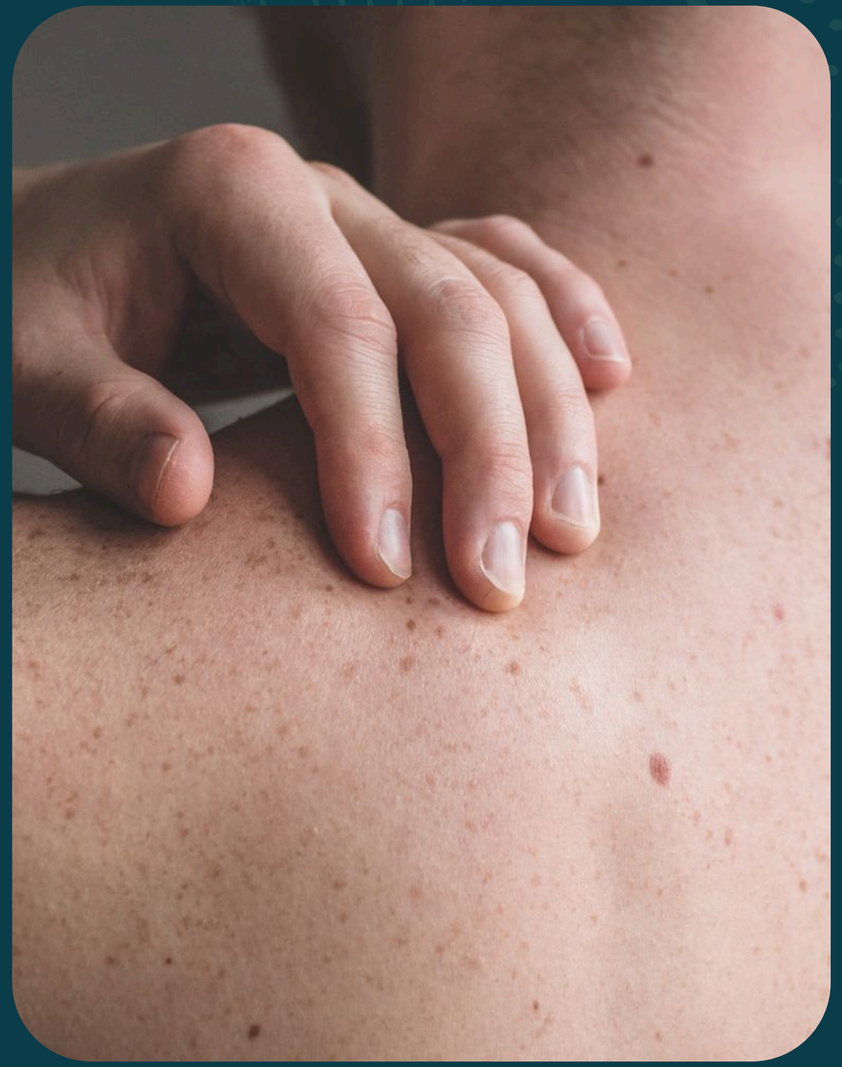
KiTH

Skin Lesion Classification

Three deep-learning approaches on a 14-class image dataset



Sokha Marady | Batch 13, Software Engineering
Lecturer: Mr. Soklong HIM | Deep Learning Final Project

Problem and motivation

Given one photo of skin, predict which of 14 classes it shows.

8

Skin lesion types

Melanoma,
melanocytic nevi,
basal cell and squamous cell carcinoma,
actinic keratoses,
benign keratosis-like,
dermatofibroma,
vascular lesions

5

Viral skin diseases

Chickenpox, cowpox, HFMD, measles, monkeypox

1

Healthy skin

Normal skin, so the model can say "nothing here"

Research question: how much does pretrained knowledge help, compared with training a network from scratch?

Dataset and split

Kaggle dataset ahmedxc4/skin-ds: 36,656 images, 14 classes, official train / val / test folders

29,322
Train (80%)

3,660
Validation (10%)

3,674
Test (10%)

- Loaded with torchvision ImageFolder from the official folders as provided; we did not re-split
- All images resized to 224 x 224; test images use the same preprocessing as validation

Heavy class imbalance

10,300

Melanocytic nevi
(train images)

about 54x

more images

191

Dermatofibroma
(train images)

Weighted cross-entropy

Rare classes count more in the loss, so the model cannot ignore them.

Macro-F1 as the main metric

Every class counts equally, unlike accuracy, which the big classes dominate.

Three approaches compared

1

Custom CNN from scratch

392,974

trainable parameters

Small CNN, random start. Learns everything from skin images only.

2

ResNet-18, frozen

7,182

trainable parameters

ImageNet weights kept fixed. Only the final linear layer is trained (linear probe).

3

ResNet-18, fine-tuned

11,183,694

trainable parameters

ImageNet weights as the starting point. Every layer is trained at a low learning rate.

Experimental setup

Setting	CNN scratch	ResNet frozen	ResNet fine-tuned
Learning rate	1e-3	1e-3	1e-4
Weight decay	1e-4	1e-4	1e-4
Max epochs	25	15	15
Training time (min)	160.6	98.0	99.9

Shared by all three

- Adam optimiser
- ReduceLROnPlateau scheduler
- Batch size 32, input 224 x 224
- Weighted cross-entropy, seed 42

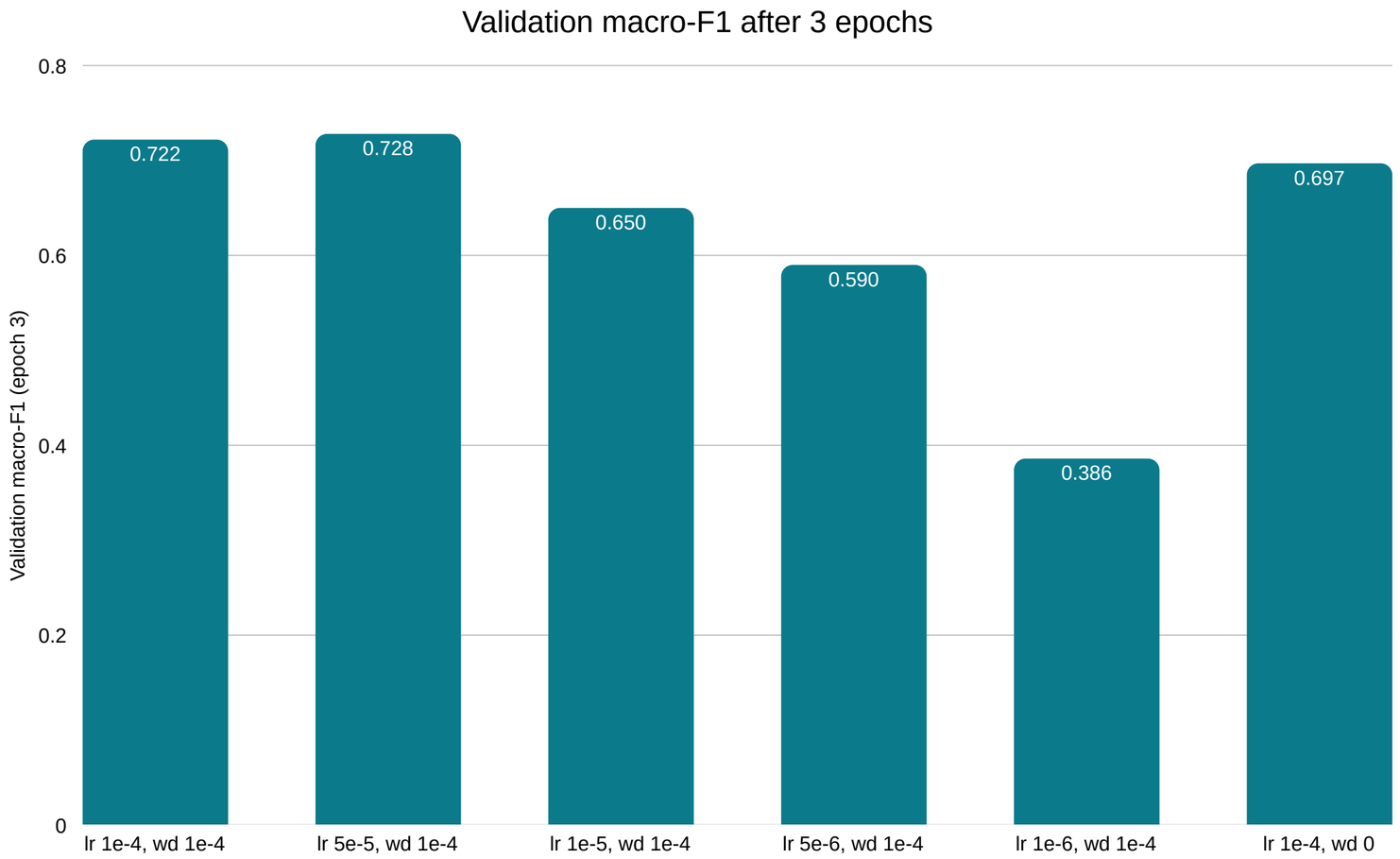
Best checkpoint

Saved at the lowest validation loss, then evaluated once on the test set.

Metrics and hardware

Accuracy, macro-F1, macro-precision, macro-recall. Tesla T4 GPU, PyTorch.

Hyperparameter tuning



What we learned

- Best: lr 5e-5 to 1e-4 (0.728 and 0.722)
- Smaller lr learned too slowly: 5e-6 = 0.590, 1e-6 = 0.386
- Removing weight decay lowered the score (0.697)
- We kept lr 1e-4; the gap to 5e-5 is within noise (single seed)

Results on the test set

Approach	Accuracy	Macro-F1	Macro-Prec.	Macro-Recall	Trainable params	Time (min)
1. CNN from scratch	0.5822	0.5205	0.5053	0.6123	392,974	160.6
2. ResNet-18 frozen	0.6184	0.5415	0.5179	0.6074	7,182	98.0
3. ResNet-18 fine-tuned	**0.8419**	**0.8330**	**0.8117**	**0.8607**	**11,183,694**	**99.9**

Fine-tuning clearly wins

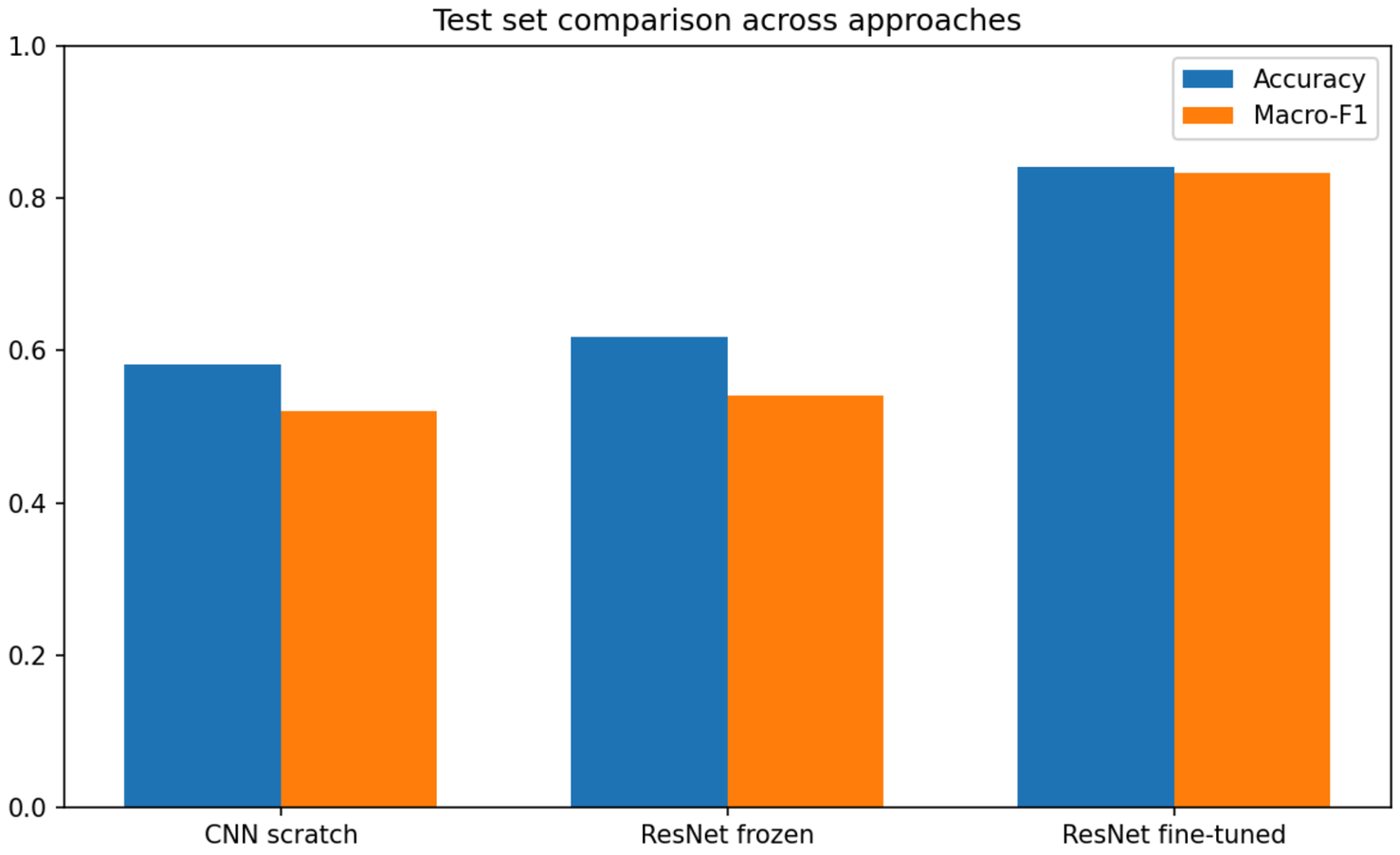
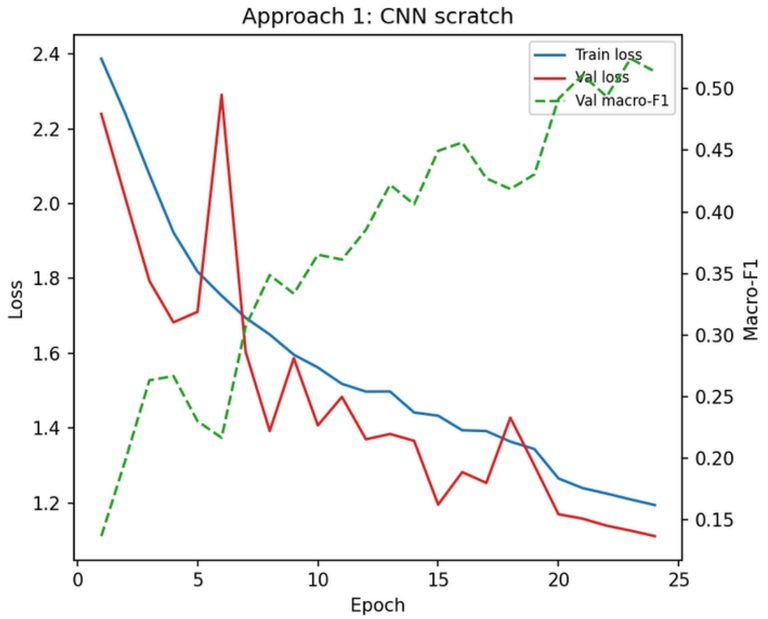
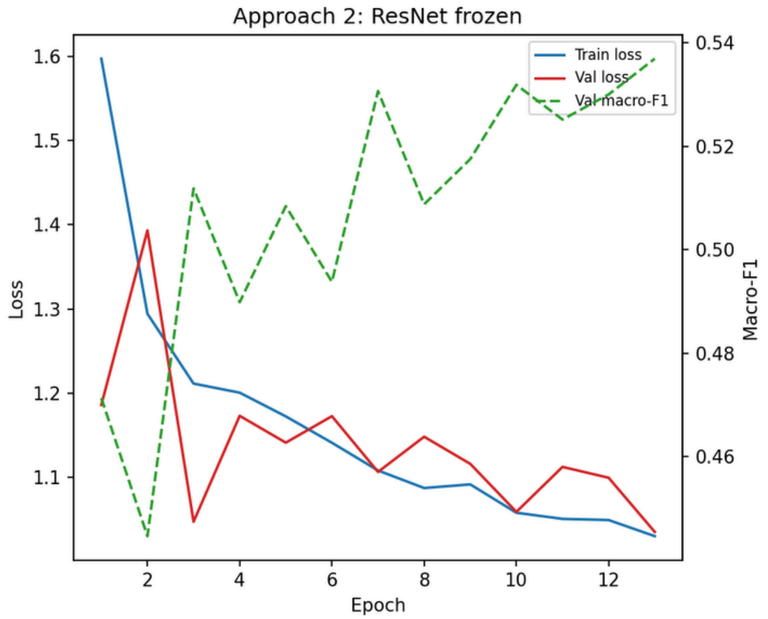


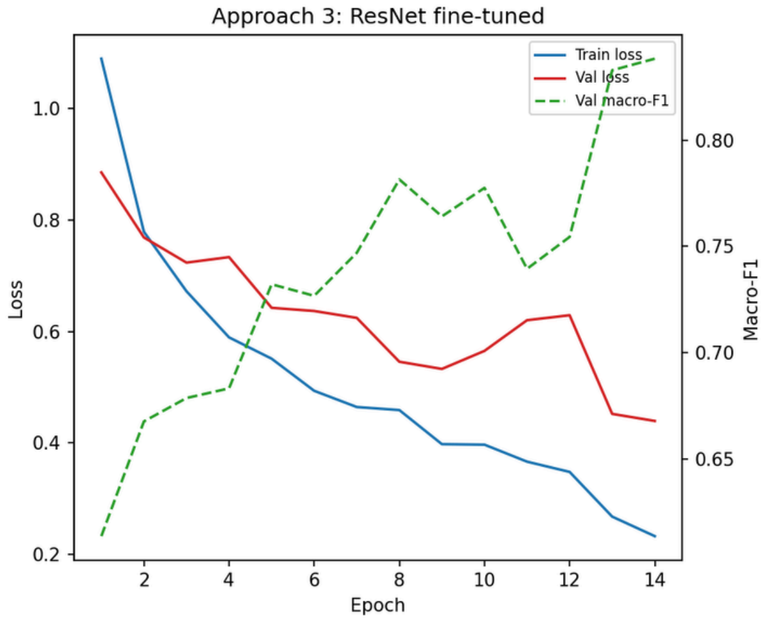
Figure: test accuracy and macro-F1

- Fine-tuned: 0.842 accuracy, 0.833 macro-F1
- Frozen and scratch models both stay near 0.6 accuracy
- The scratch CNN was the slowest to train (160.6 min)
- The fine-tuned model has a small gap between accuracy and macro-F1, so rare classes are handled better

Learning curves: overfitting and underfitting







CNN scratch: underfitting

Loss is still falling at the end, so it is undertrained or capacity-limited.

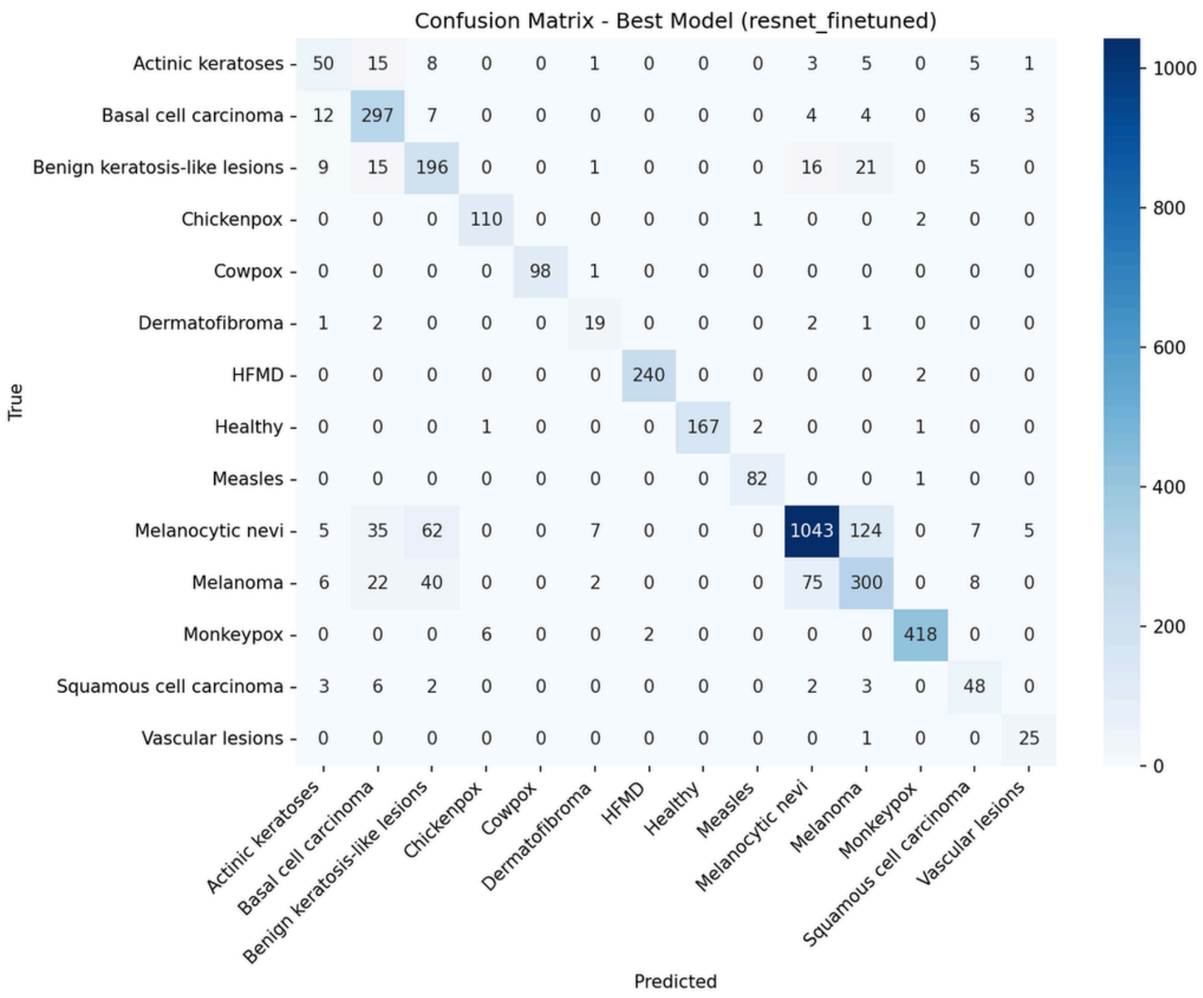
Frozen ResNet: plateau

Plateaus early with a noisy validation curve. Limited by the fixed features.

Fine-tuned: mild overfitting

Train loss 0.23 vs val loss 0.44. Val loss is still lowest at the saved best epoch.

Error analysis: confusion matrix



Nevi and melanoma mix up

124 nevi predicted as melanoma, 75 melanoma predicted as nevi

Actinic keratoses (F1 0.57)

Only 50 of 88 correct; the rest spread to basal cell, benign keratosis, SCC

Viral classes are strong

Chickenpox, cowpox, HFMD, measles, monkeypox and healthy: F1 0.96 to 0.99

Why fine-tuning won, and an informal ensembling check

Why it won

1

Frozen ImageNet features transfer poorly

Domain gap: skin photos differ from ImageNet, so a linear head on fixed features tops out at 0.618.

2

Every layer adapts to skin images

Pretrained start plus 11.2M trainable weights, versus 393k for the scratch CNN.

3

A tuned, gentle learning rate

lr 1e-4 won the tuning runs and keeps pretrained knowledge intact.

Informal ensembling check

99.62%

fine-tuned model alone, one example photo

56.68%

equal-weight ensemble of all three, same photo

Here the weak CNN diluted the strong model. Not tested on the full test set.

Limitations and future work

Limitations

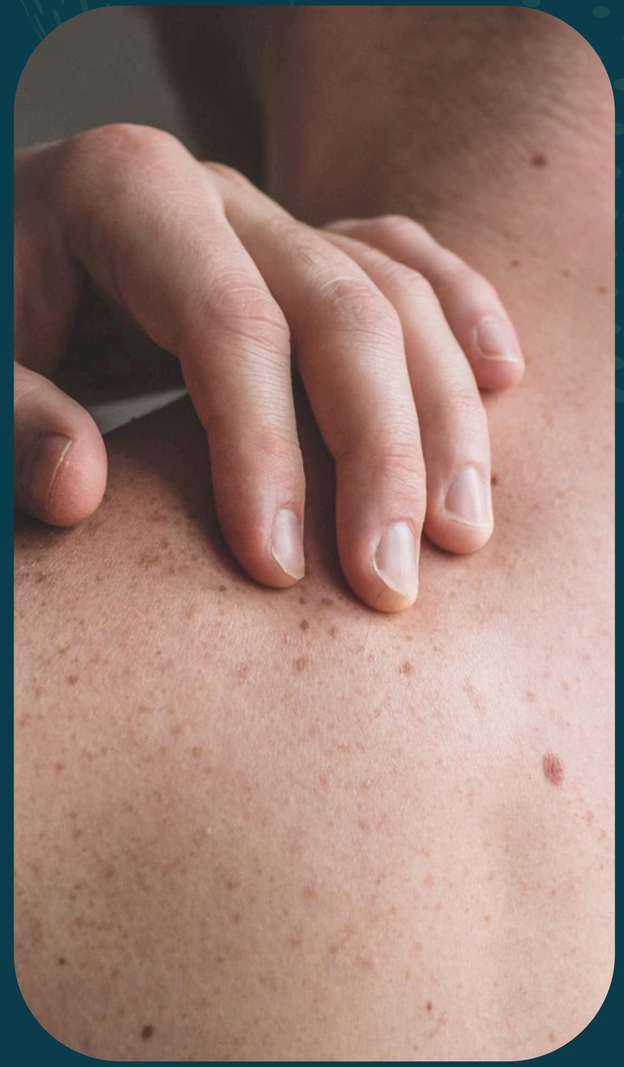
- Domain shift: can fail on random internet photos (informal tests), since it learned one dataset's photo style
- Multi-source acquisition bias: image source can leak into what the model learns
- Rare, look-alike classes stay weak: actinic keratoses, melanoma, dermatofibroma, SCC
- Not a diagnostic tool

Future work

- Data augmentation and more samples for rare classes
- Weighted or learned ensembling instead of equal weights
- Test on external datasets to measure domain shift
- Try larger pretrained backbones

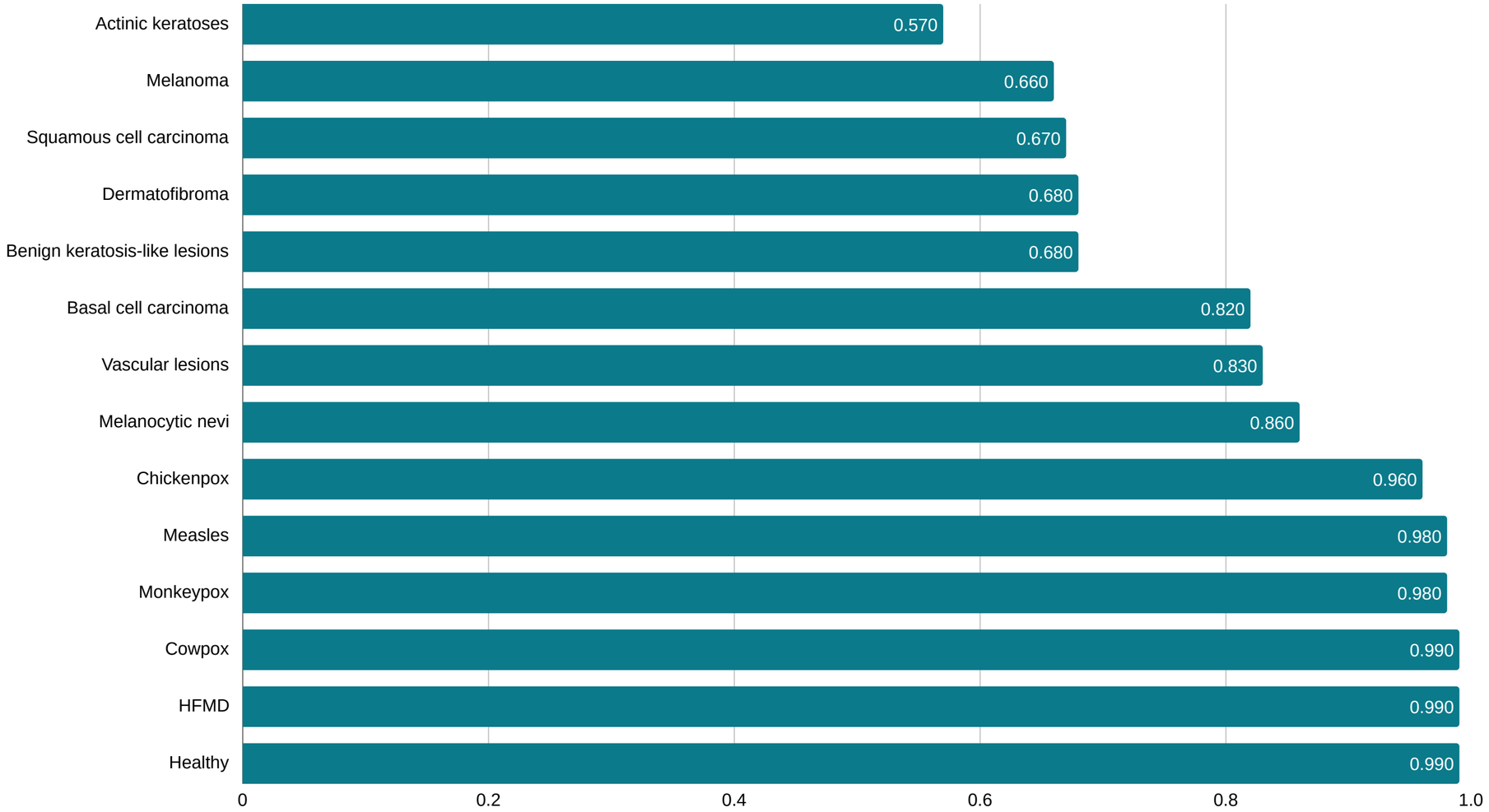
Conclusion

- Fully fine-tuned ResNet-18 reached 0.842 accuracy and 0.833 macro-F1
- Frozen features and a scratch CNN both stayed near 0.6 accuracy
- Main errors: nevi vs melanoma and other look-alike lesions
- Next: rare-class data and external validation



github.com/Smarady67/skin-lesion-classification

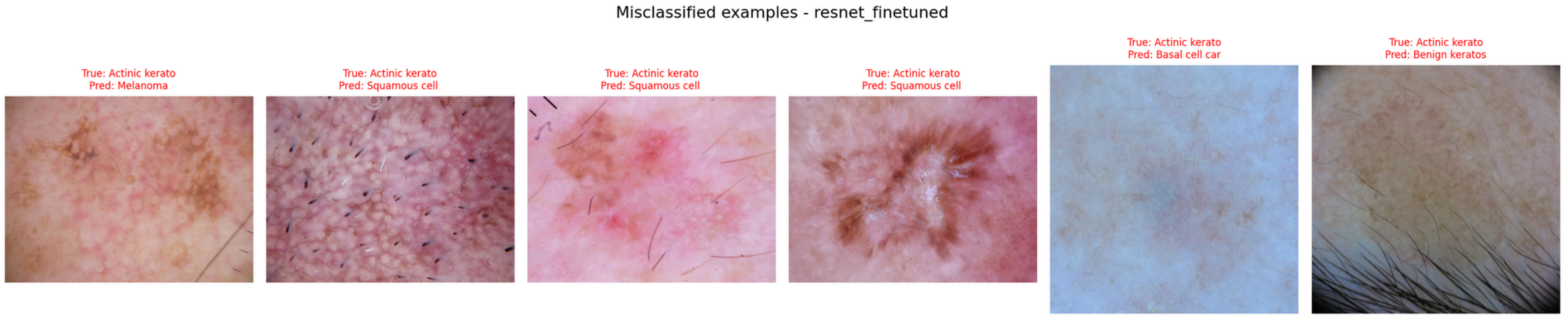
Appendix A: per-class F1, fine-tuned ResNet



Reading the chart

- Coral: F1 below 0.70
- Weak classes are rare and look alike
- Viral classes: 0.96 to 0.99

Appendix B: misclassified examples



- All six shown are actinic keratoses, predicted as melanoma, squamous cell carcinoma, basal cell carcinoma or benign keratosis
- These classes share colour and texture, so even a strong model confuses them